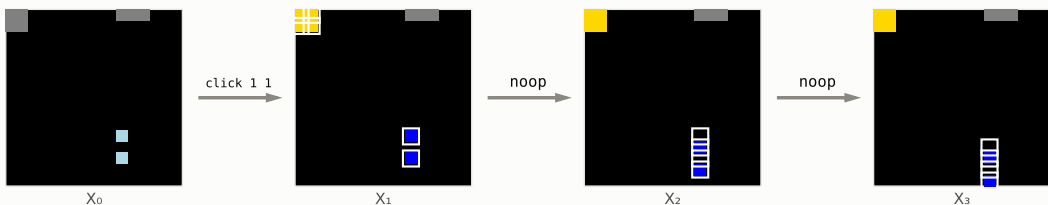


Ice (BT3GB) — three transitions from one training drive



$X_0 \rightarrow X_1$ **click 1 1** one click toggles two things at once: the 2x2 latch gray→gold, and both water cells lightblue→blue
 $X_1 \rightarrow X_2$ **noop** no action, and each water cell falls one row
 $X_2 \rightarrow X_3$ **noop** and again — the lower one reaches the floor at row 15

	abstraction program P	world knowledge K
node 0 seed 0.00 min(ID, cFD)	blank slate — parses the grid, emits nothing <pre> grid = _parse_grid(obs) ... return "" P(X₁) = (empty string) P(X₂): identical — the transition is invisible to P </pre>	(empty — no world knowledge yet)
node 3 iter 3 : P 0.10 min(ID, cFD)	updated — every non-background cell, one by one <pre> cells.append((r, c, colour)) ... cell_strs = [f"{{colour}}({r}),{{c}})" for r, c, colour in cells] parts.append(", ".join(cell_strs)) P(X₁) = bg=black; gold(0,0), gold(0,1), gray(0,10), gray(0,11), gray(0,12), gold(1,0), gold(1,1), blue(11,10), blue(13,10) P(X₂): -blue(11,10) -blue(13,10) +blue(12,10) +blue(14,10) </pre>	(empty — no world knowledge yet)
node 4 iter 4 : K 0.53 min(ID, cFD)	unchanged from node 3	new - There is a fixed 2x2 block at (0,0),(0,1),(1,0),(1,1) that toggles between gold and gray - There is a contiguous block of three gray cells on row 0 (the... movable block ...) that moves left/right with the corresponding direction actions - On click (any ROW COL): all cells in the top-left 2x2 block toggle gold↔gray; all blue/lightblue cells anywhere on the grid toggle blue↔lightblue
node 12 iter 22 : P 0.73 min(ID, cFD)	rewritten at node 8, refined here — it emits the objects K names <pre> gray_row0_cols = sorted([c for r,c,col in cells if r==0 and col=="gray" and c not in fixed_cols_row0]) ... parts.append("fixed:" + ", ".join(fixed_entries)) parts.append(f"movable:gray_row0:{{movable_run[0]}}-{{ movable_run[1]}}") P(X₁) = step=0; bg=black; fixed:gold(0,0),gold(0,1),gold(1,0),gold(1,1); movable:gray_row0:10-12; other:blue_row11:10; blue_row13:10 P(X₂): -blue_row11:10 -blue_row13:10 +blue_row12:10 +blue_row14:10 </pre>	revised at node 6, then frozen - These cells are either gold or gray. Their color persists even when covered by the movable block; they are visible only when not covered - On noop: every blue/lightblue cell not on row 15 attempts to move down one row. If the cell directly below is empty... it moves - If ROW in {0,1} and COL in {0,1} (i.e. clicked inside the top-left 2x2 block), then the fixed 2x2 block toggles... Else (click outside the top-left 2x2 block), then both of the following occur ↑ $X_0 \rightarrow X_1$ is a click at (1,1) and the water toggles too — node 4 had this rule right